

Select and Configure Compute — Knowledge Check

20 questions · Select and configure compute in Azure Databricks (Microsoft Learn, Unit 4). Answer key starts on a new page.

1. What is the typical startup time for serverless compute?
 - A. 2-6 seconds
 - B. 3-7 minutes
 - C. 4-6 minutes
 - D. Under 1 minute
2. Which serverless job/pipeline performance mode can reduce DBU consumption by up to 70%, at the cost of a 4-6 minute startup?
 - A. Performance-optimized
 - B. Standard
 - C. Burst
 - D. Economy
3. True or False: Serverless compute in notebooks supports Scala, R, JAR libraries, and RDD APIs.

4. Which classic compute access mode assigns a cluster exclusively to a single user or group and is required for RDD APIs, GPU workloads, R, or custom containers?
 - A. Standard access mode
 - B. Dedicated access mode
 - C. No isolation shared
 - D. Auto access mode
5. What's the key architectural limitation of a single-node cluster?
 - A. It can't run Python
 - B. It has no worker nodes, so it can't distribute computation across machines
 - C. It can't be used for machine learning
 - D. It requires Unity Catalog
6. Which SQL warehouse type supports Intelligent Workload Management and starts in 2-6 seconds?
 - A. Classic
 - B. Pro
 - C. Serverless
 - D. All three equally
7. What cost do you pay for an idle instance sitting in an instance pool?
 - A. Nothing at all
 - B. Virtual machine costs, but not Databricks compute units (DBUs)
 - C. Full DBU rate plus VM cost
 - D. Only a flat monthly fee
8. What Azure Databricks error can occur when your cloud provider runs out of capacity for your specified instance type?

9. How often does optimized autoscaling evaluate utilization for job compute versus all-purpose compute?

- A. Every 10 seconds for both
- B. Every 40 seconds for job compute, every 150 seconds for all-purpose compute
- C. Every 150 seconds for job compute, every 40 seconds for all-purpose compute
- D. Once per hour for both

10. True or False: You should use spot instances for driver nodes to save on cost.

11. Which node/instance family is best suited for workloads with large joins and aggregations that need to stay in memory?

- A. Compute-optimized (F-series)
- B. Memory-optimized (E-series)
- C. Storage-optimized (L-series)
- D. GPU-accelerated (NC/ND-series)

12. On which Databricks Runtime version is Photon enabled by default?

- A. 9.1 LTS and above
- B. Only Runtime ML versions
- C. 13.3 LTS and above only
- D. Photon is never on by default

13. For production job compute running daily transformations, which type of Databricks Runtime version is recommended for stability?

- A. The absolute newest runtime available
- B. Long Term Support (LTS) versions
- C. Beta/preview runtimes
- D. Runtime version doesn't matter

14. What permission must you have on a cluster to install compute-scoped libraries?

15. On standard access mode clusters, what must happen before Maven coordinates or JAR files can be installed?

- A. Nothing extra is required
- B. A metastore admin must add them to an allowlist
- C. They must be converted to Python wheels
- D. The cluster must be restarted twice

16. What's the recommended use for init scripts, given that Databricks doesn't recommend them for library installation?

- A. Installing Python packages from PyPI
- B. System-level configuration like apt-get packages, environment variables, or monitoring agents
- C. Creating Unity Catalog schemas
- D. Managing compute permissions

17. Which compute permission level allows a user to attach notebooks and view the Spark UI, but not start, stop, or restart the compute?

- A. NO PERMISSIONS
- B. CAN ATTACH TO

- C. CAN RESTART
- D. CAN MANAGE

18. Why is 'No isolation shared' access mode considered legacy and discouraged?

- A. It's too expensive
- B. Users with only CAN ATTACH TO permission can view service account keys in log4j files
- C. It doesn't support Python
- D. It requires GPU instances

19. What is required to create a compute resource dedicated to a specific group (formerly 'Single-user' access)?

- A. Unity Catalog enabled and Databricks Runtime 15.4 or above
- B. Just workspace admin rights
- C. A classic SQL warehouse
- D. Nothing special — it's the default

20. On a group-dedicated cluster, who becomes the owner of a data object (e.g. a schema) created by a group member?

- A. The individual user who ran the command
- B. The group itself
- C. The workspace admin
- D. No owner is assigned

Answer Key

1. A. 2-6 seconds

Serverless compute is fully managed by Databricks and maintains warm capacity, enabling near-instant startup — far faster than classic compute's 3-7 minutes.

2. B. Standard

Standard mode is for scheduled batch jobs/pipelines where startup latency is acceptable; it's not available for notebooks, continuous pipelines, or one-time runs/submit calls.

3. False

Serverless notebooks don't support Scala, R, JAR libraries, or RDD APIs, and custom Spark configs are restricted. Use classic compute with dedicated access mode for these.

4. B. Dedicated access mode

Dedicated access mode grants full machine-level privileges, unlike Standard mode which uses Lakeguard to isolate multiple concurrent users on one shared cluster.

5. B. It has no worker nodes, so it can't distribute computation across machines

Single-node clusters (driver only) suit lightweight exploration, small datasets, or ML frameworks like scikit-learn that don't inherently distribute — not workloads needing horizontal scaling.

6. C. Serverless

Serverless SQL warehouses combine Photon, Predictive IO, and Intelligent Workload Management for the fastest, most dynamic scaling of the three types.

7. B. Virtual machine costs, but not Databricks compute units (DBUs)

This is the tradeoff: you pay to keep VMs warm and ready, but save on DBU charges, which is worthwhile when frequent cluster creation justifies the idle cost.

8. CLOUD_PROVIDER_RESOURCE_STOCKOUT

Flexible node types solve this by automatically falling back to compatible alternative instance types (matching vCPU, memory within 100-110%, disk, architecture, and OS image support).

9. B. Every 40 seconds for job compute, every 150 seconds for all-purpose compute

Optimized autoscaling (the default) also scales up in two quick steps to the maximum and can scale down even when not fully idle, based on shuffle file state.

10. False

Spot instances are fine for worker nodes since Spark tolerates their failure, but the driver should always run on an on-demand instance — losing the driver kills the entire cluster.

11. B. Memory-optimized (E-series)

Memory-optimized instances provide a higher RAM-to-core ratio, reducing the chance of spilling to disk during memory-intensive operations.

12. A. 9.1 LTS and above

Photon accelerates SQL and DataFrame operations with native code, but it is not supported on GPU-enabled clusters — it must be disabled for GPU/ML workloads.

13. B. Long Term Support (LTS) versions

LTS runtimes provide extended compatibility so you can thoroughly test before upgrading, avoiding unexpected disruption to production pipelines.

14. CAN MANAGE

CAN MANAGE allows modifying cluster configuration, including adding and removing libraries; without it, you can't access the library installation interface.

15. B. A metastore admin must add them to an allowlist

This security measure ensures admins review and approve libraries running on shared compute; allowlisting also uses prefix matching for paths (e.g. a trailing slash matters).

16. B. System-level configuration like apt-get packages, environment variables, or monitoring agents

Cluster-scoped libraries are the better approach for installing packages; init scripts run before Spark starts and fail the cluster if any script returns a non-zero exit code.

17. B. CAN ATTACH TO

CAN ATTACH TO is the base usable permission level; lifecycle control (start/stop/restart) requires at least CAN RESTART.

18. B. Users with only CAN ATTACH TO permission can view service account keys in log4j files

This is a real security exposure. Standard or Dedicated access modes provide proper isolation and should be used instead.

19. A. Unity Catalog enabled and Databricks Runtime 15.4 or above

The assigned group also needs CAN MANAGE permission on a workspace folder for storing notebooks and experiments used on that cluster.

20. B. The group itself

Individual permissions scope down to the group's permissions on a dedicated group cluster, so objects created there are owned by the group, not the individual.